Sharp server heuristic: influence and finite-breakdown characterization

The server-side `robust_aggregate` rule is the sharp heuristic axis of the three-axes design (sec. 10.2.1).

Sharp server heuristic (part 2)

It has BH-rejected positive contrasts in the configured accuracy verdict in sec. 10.5.2 but has declared reversals elsewhere.

Sharp server heuristic (part 3)

Unlike the objective-backed `variational_aggregate`, no closed-form free-energy derivation has been established for it in this repository. A separate scoped proposition in the aggregation-objective supplement rules out the declared continuously differentiable,

Sharp server heuristic (part 4)

separable forward-KL objective class for the implementation's raw log-pool block; it does not rule out every broader coupled or fixed-point-only construction.

Sharp server heuristic (part 5)

The rule therefore remains a heuristic whose positive formal property is bit-identical recovery of the log-linear pool at `robustness = 0` (eq. 10). This section does not promote the scoped negative result into an objective certificate; it *measures* the heuristic empirically,

Sharp server heuristic (part 6)

and the measurement makes its honesty boundary concrete.

Sharp server heuristic (part 7)

We measure two things (fig. 36). First, a **numerical influence function**: we drag one agent's belief a growing fraction toward a confident-wrong contamination point and read its converged pooling weight. At `robustness = 0` the weight is a flat $1/n$ at every perturbation —

Sharp server heuristic (part 8)

the naive pool never down-weights anyone — which anchors the instrument to the proven recovery corner.

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Sharp server heuristic (part 9)

At positive robustness the dragged agent's influence falls (not strictly monotonically — a tiny drag can briefly *raise* it before the divergence penalty dominates, an honest non-monotonicity we report rather than smooth away).

Sharp server heuristic (part 10)

Second, and more consequentially, a **breakdown witness**. We add colluding confident-wrong adversaries — all broadcasting the same false state — to a fixed colony of 5 honest sentinels until each aggregator's consensus argmax is *captured* (flips to the adversaries' target). The sharp heuristic is captured by 2 colluders;

Sharp server heuristic (part 11)

the conservative objective-backed variational rule withstands more, capitulating only at 4.

Sharp server heuristic (part 12)

Both counts are **finite** (Yes): a colluding majority overwhelms either rule. That finite breakdown point is the honest headline: neither rule has an unconditional truth-recovery claim under coordinated collusion. The absence of an objective theorem for `robust_aggregate` is a separate derivational boundary.

Sharp server heuristic (part 13)

The finite capture measurement neither establishes estimator-level B-robustness nor refutes the variational rule's stated raw effective-weight result.

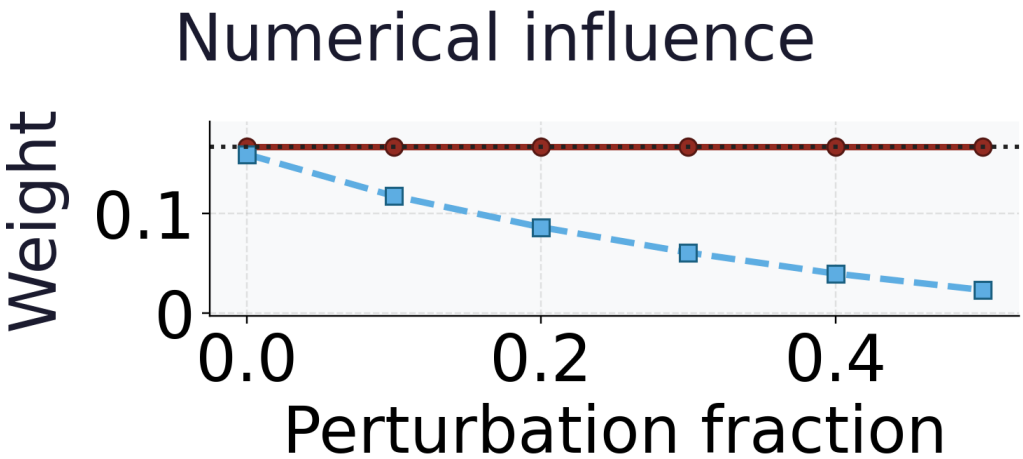
Sharp server heuristic (part 14)

The report also runs a declared diagnostic grid over state dimension, honest-agent count, robustness, four simple attack mechanisms, and balanced versus adversary-downweighted base weights. This is a coverage instrument for finding counterexamples,

Sharp server heuristic (part 15)

not a random sample of worlds and not a theorem search over all simplexes. A finite capture row is evidence against a universal guarantee; an uncaptured row is only “not found within this search budget.”

Sharp server heuristic (part 16)

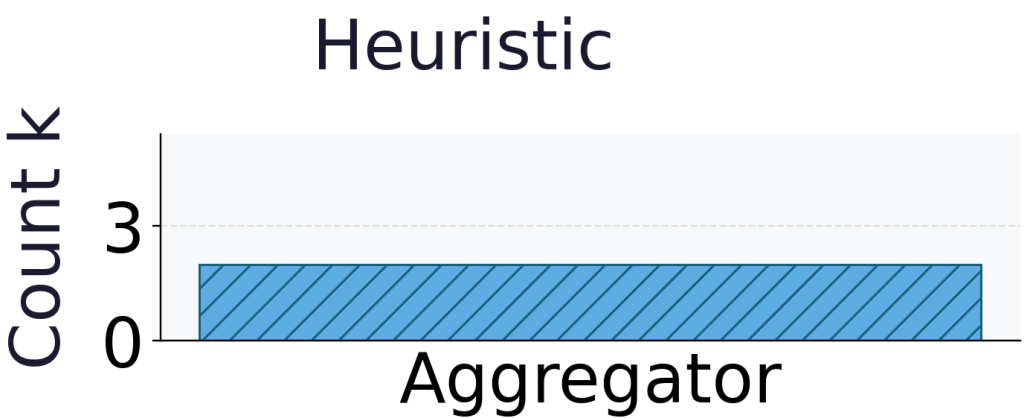


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Sharp server heuristic (part 17)

Endpoint weight drop: reference minus heuristic = 0.1435. Empirical at these settings, not a guarantee.

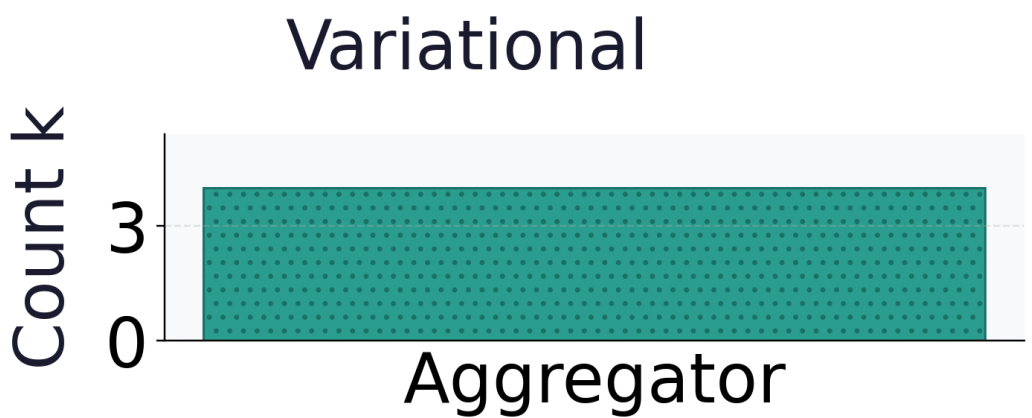
Sharp server heuristic (part 18)



Sharp server heuristic (part 19)

Heuristic: argmax captured at k=2.

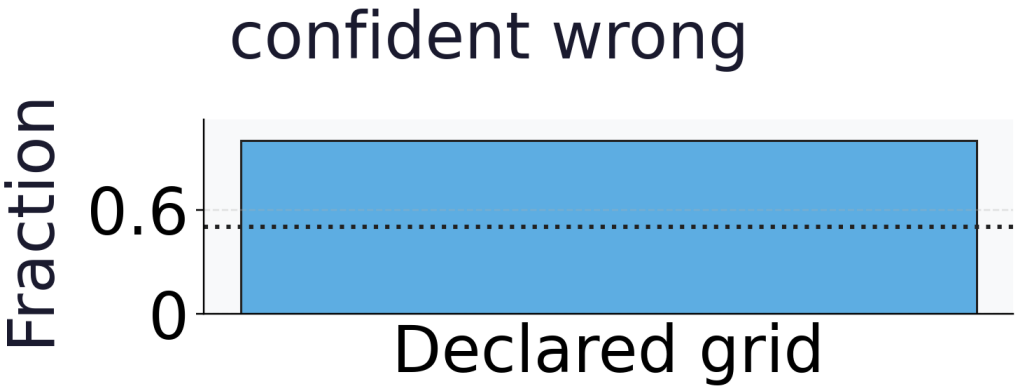
Sharp server heuristic (part 20)



Sharp server heuristic (part 21)

Variational: argmax captured at k=4.

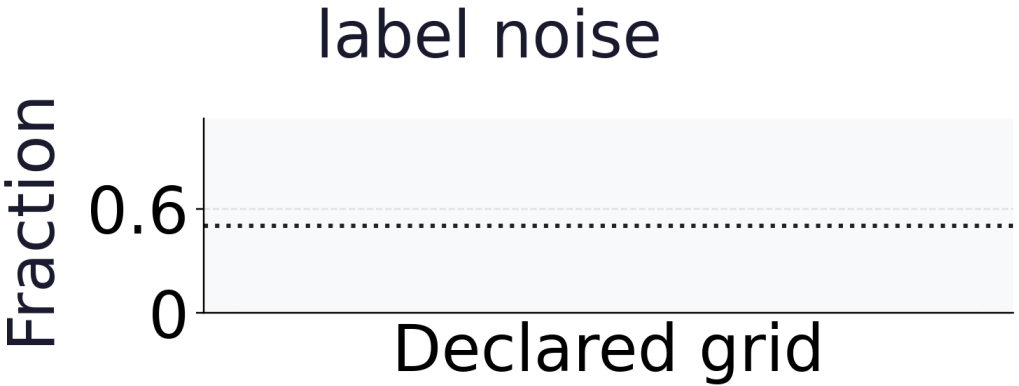
Sharp server heuristic (part 22)



Sharp server heuristic (part 23)

confident wrong: 16/16 rows with finite capture = 100%.

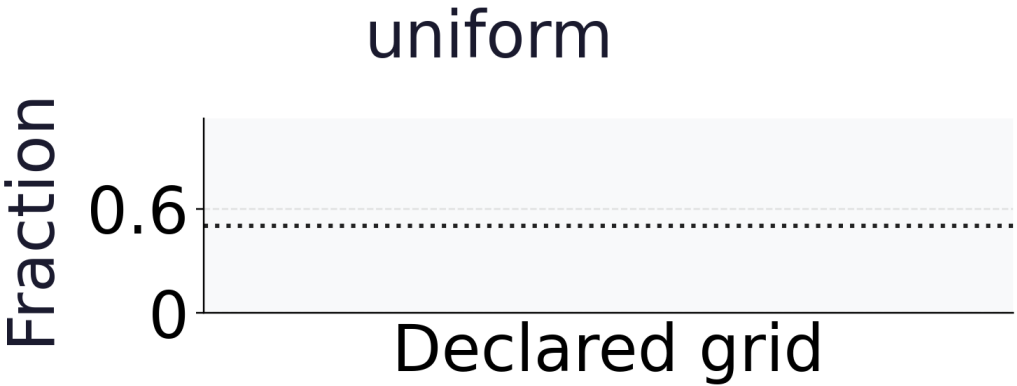
Sharp server heuristic (part 24)



Sharp server heuristic (part 25)

label noise: 0/16 rows with finite capture = 0%.

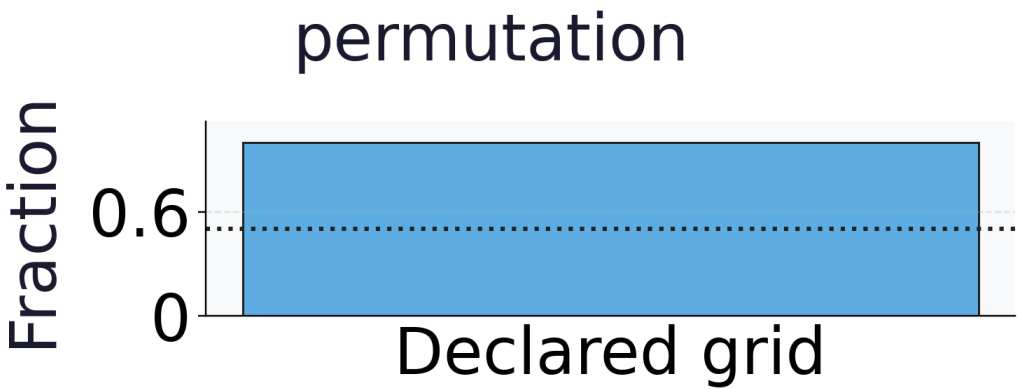
Sharp server heuristic (part 26)



Sharp server heuristic (part 27)

uniform: 0/16 rows with finite capture = 0%.

Sharp server heuristic (part 28)



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Sharp server heuristic (part 29)

permutation: 16/16 rows with finite capture = 100%.

Sharp server heuristic (part 30)

Finite-search capture and grid frequencies describe the declared settings. They are not global breakdown probabilities or universal robustness guarantees.